

WT_24ITPW501

1. Code Editor (Essential)

- [Visual Studio Code](#) – Recommended because it supports HTML, CSS, JavaScript, Node.js, database extensions, and debugging in one place.

Useful extensions in VS Code:

- Live Server
- Prettier
- ESLint
- MongoDB for VS Code
- REST Client

2. Web Browsers

Install at least two browsers to demonstrate compatibility and developer tools:

- [Google Chrome](#)
- [Mozilla Firefox](#)

Their developer tools are excellent for teaching HTML, CSS, JavaScript, AJAX, and network requests.

3. Runtime for Server-Side Programming

- [Node.js](#)

Installing Node.js automatically installs **npm (Node Package Manager)**, which is needed to install libraries such as Express.

4. Backend Framework

- [Express.js](#)

This will be used in Module V for REST APIs and server-side development.

5. Database Software

Choose one or teach both:

- [MongoDB Community Server](#)
- [MySQL Community Server](#)

For beginners, MongoDB integrates very smoothly with Node.js and Express.

6. Database GUI Tools

- [MongoDB Compass](#) for MongoDB
- [MySQL Workbench](#) for MySQL

These help students visualize tables, collections, and queries.

7. API Testing Tool

- [Postman](#)

Useful for testing REST APIs in Module V.

8. Optional Networking Tools

For Module VI (HTTP, FTP, DNS, SSH, SNMP, sockets):

- [Wireshark](#) – Packet analysis and protocol demonstrations.
 - [PuTTY](#) – Demonstrating SSH and Telnet.
 - Built-in terminal (ping, nslookup, tracer, netstat) in Windows/Linux.
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Minimum Installation Set for a Lab Course

If you want the simplest setup for students:

1. Visual Studio Code
2. Google Chrome
3. Node.js (includes npm)
4. MongoDB Community Server
5. MongoDB Compass
6. Postman

This combination is sufficient to cover almost the entire syllabus.

Ideal Laboratory Configuration

For a 45-period undergraduate Web Technology course:

- Windows 10/11 or Ubuntu Linux
- 8 GB RAM minimum (16 GB preferred)
- VS Code
- Chrome + Firefox
- Node.js + Express
- MongoDB + Compass
- Postman
- Wireshark

That setup comfortably supports all modules from HTML basics through REST APIs and networking concepts